TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	78.6 ± 2.80
Correlation Stability	88.0 ± 2.14
Distribution Consistency	85.6 ± 1.91
Entropy Preservation	87.9 ± 0.70
Mean Z Score Drift	0.2 ± 0.03
Categorical Fidelity	83.2 ± 0.60
Privacy Risk	49.3 ± 1.55
Data Density	99.9 ± 0.20
Coverage Score	21.5 ± 4.20
Feature Accuracy	78.6 ± 1.80
Novelty Score	0.5 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	42.356	-2.390	5.3%	OK
TB	3.299	1.383	-1.916	58.1%	Drift
DB	1.486	0.542	-0.944	63.6%	Drift
Alkphos	290.576	225.955	-64.621	22.2%	Warn
Sgpt	80.714	42.054	-38.660	47.9%	Drift
Sgot	109.911	41.021	-68.889	62.7%	Drift
TP	6.483	6.716	+0.232	3.6%	OK
ALB	3.142	3.523	+0.381	12.1%	Warn
A/G Ratio	0.947	1.059	+0.112	11.8%	Warn
Selector	1.286	1.354	+0.068	5.3%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	78.6 ± 2.80
Correlation Stability (*2)	88.0 ± 2.14
Distribution Consistency (*3)	85.6 ± 1.91
Entropy Preservation (*4)	87.9 ± 0.70
Mean Z Score Drift (*5)	0.2 ± 0.03
Categorical Fidelity (*6)	83.2 ± 0.60
Privacy Risk (*7)	49.3 ± 1.55
Data Density (*8)	99.9 ± 0.20
Coverage Score (*9)	21.5 ± 4.20

Feature Accuracy (*10)	78.6 ± 1.80
Novelty Score (*11)	0.5 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.5470085470085471	0.5632478632478632	0.00427350427350429	
F1	0.5587432119380917	0.6403357266972324	0.5697780265818274	0.01103481464373568	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6213675213675214	0.6854700854700855	-0.00085470085470085	
F1	0.7971299319768493	0.7476226950966753	0.7995487155898496	0.00241878361300024	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6820512820512821	0.5316239316239317	0.7059829059829059	0.02393162393162393	*
F1	0.7891274144551341	0.6378185977895721	0.8068120331516531	0.01768461869651893	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6598290598290599	0.5435897435897437	0.6495726495726496	-0.01025641025641025	
F1	0.7624245828764089	0.6525503166311617	0.76461277159156	0.00218818871515098	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	0.5726495726495726	0.7102564102564102	0.00683760683760685	
F1	0.8108734553482932	0.6940120106185133	0.8187418911238847	0.00786843577559140	

Model: XGB

TabuLLM: AI Benchmark Report

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.5589743589743592	0.7042735042735042	0.0017094017094017033	
F1	0.7997903571549948	0.6675625839863379	0.8010443637777878	0.0012540066227930136	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6683760683760684	0.7068376068376068	0	
F1	0.8279091624529537	0.7974672661547826	0.8279091624529537	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.5743589743589743	0.6897435897435897	0.015384615384615396	
F1	0.7821343746683586	0.692422040185973	0.8011793955282845	0.019045020859925977	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6957264957264957	0.5547008547008547	0.6675213675213676	-0.028205128205128206	
F1	0.7989086216255812	0.6628997795206433	0.785595431327158	-0.013313190298423183	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7102564102564103	0.5786324786324786	0.7119658119658119	0.0017094017094017144	
F1	0.8129113973301338	0.6984694833874777	0.8168951752251067	0.003983777894972751	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Female	0.9	0.2	165	26	29	8.5	4.4	1.0
2	43	Male	1.3	0.6	155	15	20	8.0	4.0	1.0
3	42	Male	16.4	8.9	245	56	87	5.4	2.0	0.5
4	18	Male	0.9	0.3	300	30	48	8.0	4.0	1.0
5	33	Male	0.8	0.2	135	30	29	7.2	4.4	1.5
6	36	Female	1.2	0.4	358	160	90	8.3	4.4	1.1
7	29	Female	1.0	0.4	210	35	40	6.8	3.5	1.2
8	45	Male	0.9	0.3	180	25	30	7.0	3.8	1.1
9	52	Female	2.5	1.0	250	45	50	6.5	3.2	1.0
10	37	Male	1.2	0.5	190	30	35	7.2	3.6	1.3
11	41	Female	0.8	0.2	160	28	25	6.7	3.4	1.0
12	34	Male	0.7	0.2	175	22	28	6.9	3.1	0.9
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	2.0	1.1	664	52	104	6.0	2.1	0.53
2	16	Male	0.7	0.2	418	28	35	7.2	4.1	1.3
3	35	Male	1.8	0.6	275	48	178	6.5	3.2	0.9
4	38	Male	0.8	0.2	247	55	92	7.4	4.3	1.38
5	54	Male	2.2	1.2	195	55	95	6.0	3.7	1.6
6	55	Male	18.4	8.8	206	64	178	6.2	1.8	0.4
7	25	Female	1.1	0.4	220	30	45	6.5	3.2	1.0
8	34	Male	0.9	0.3	180	40	50	7.0	3.8	1.1
9	42	Female	2.5	1.0	300	50	60	7.5	4.0	1.2
10	29	Male	0.8	0.2	200	35	48	6.8	3.5	1.1
11	55	Female	1.3	0.5	250	45	55	7.2	3.9	1.0
12	38	Male	0.7	0.1	150	32	40	6.2	3.0	0.8
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	21	Male	0.7	0.2	211	14	23	7.3	4.1	1.2
2	58	Female	2.8	1.3	670	48	79	4.7	1.6	0.5
3	32	Male	0.7	0.2	165	31	29	6.1	3.0	0.96
4	70	Male	3.1	1.6	198	40	28	5.6	2.0	0.5
5	55	Male	0.9	0.2	116	36	16	6.2	3.2	1.0
6	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
7	29	Female	1.2	0.4	210	22	30	6.5	3.2	1.0
8	34	Male	0.8	0.3	180	25	35	7.0	3.5	1.1
9	50	Female	2.5	1.0	300	40	50	6.0	2.5	0.83
10	45	Male	1.5	0.6	250	30	45	7.2	3.8	1.2
11	39	Female	0.9	0.2	220	28	32	6.8	3.4	1.0
12	27	Male	1.1	0.5	190	24	29	7.1	3.6	1.2
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
2	72	Male	0.7	0.1	196	20	35	5.8	2.0	0.5
3	32	Male	3.7	1.6	612	50	88	6.2	1.9	0.4
4	33	Male	0.8	0.2	135	30	29	7.2	4.4	1.5
5	37	Female	0.8	0.2	205	31	36	9.2	4.6	1.0
6	49	Female	0.6	0.1	185	17	26	6.6	2.9	0.7
7	29	Female	1.5	0.5	210	30	40	6.5	3.0	0.8
8	52	Male	0.9	0.2	150	20	25	5.8	2.5	0.6
9	41	Female	2.2	1.0	300	50	60	7.0	3.4	1.0
10	34	Male	1.1	0.3	220	35	45	6.2	3.1	0.9
11	60	Female	3.5	1.5	400	80	90	7.5	3.8	1.2
12	27	Male	0.7	0.1	180	25	30	5.5	2.8	0.7
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	21	Male	0.8	0.2	183	33	57	6.8	3.5	1.0
2	38	Female	0.8	0.2	145	19	23	6.1	3.1	1.03
3	50	Male	7.3	3.6	1580	88	64	5.6	2.3	0.6
4	75	Female	0.8	0.2	205	27	24	4.4	2.0	0.8
5	48	Male	5.0	2.6	555	284	190	6.5	3.3	1.0
6	61	Male	0.8	0.2	163	18	19	6.3	2.8	0.8
7	22	Female	0.9	0.3	210	25	30	6.5	3.2	1.1
8	34	Male	1.5	0.7	320	40	50	7.0	3.8	1.0
9	29	Female	0.8	0.2	180	20	25	6.0	3.0	1.0
10	45	Male	2.0	1.0	400	50	60	6.8	3.4	1.0
11	55	Female	0.7	0.1	250	30	35	7.2	3.5	1.2
12	60	Male	3.5	1.5	500	70	80	7.5	3.9	1.0
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	26	Male	0.6	0.2	120	45	51	7.9	4.0	1.0
2	25	Male	0.7	0.2	185	196	401	6.5	3.9	1.5
3	45	Male	2.4	1.1	168	33	50	5.1	2.6	1.0
4	39	Male	1.9	0.9	180	42	62	7.4	4.3	1.38
5	53	Female	0.7	0.1	182	20	33	4.8	1.9	0.6
6	60	Male	6.3	3.2	314	118	114	6.6	3.7	1.27
7	34	Female	1.2	0.4	210	45	60	7.5	4.0	1.2
8	29	Male	0.9	0.3	150	30	35	6.8	3.5	1.0
9	42	Female	0.6	0.2	175	25	30	6.0	3.2	0.9
10	55	Male	2.5	1.1	300	100	120	8.0	4.5	1.1
11	38	Female	1.1	0.5	220	40	50	7.0	3.8	0.95
12	45	Male	0.8	0.2	180	35	40	6.5	3.0	1.0
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	33	Male	0.7	0.1	168	35	33	7.0	3.7	1.1
2	37	Male	0.8	0.2	147	27	46	5.0	2.5	1.0
3	50	Male	0.7	0.2	188	12	14	7.0	3.4	0.9
4	58	Male	1.7	0.8	188	60	84	5.9	3.5	1.4
5	47	Male	0.9	0.2	192	38	24	7.3	4.3	1.4
6	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8
7	34	Male	0.9	0.3	175	40	30	6.5	3.6	1.0
8	29	Female	1.2	0.4	220	45	50	7.2	4.0	1.1
9	41	Male	0.8	0.2	190	35	40	6.8	3.8	1.2
10	55	Female	1.5	0.5	300	60	70	8.0	4.5	1.3
11	38	Male	2.0	0.6	250	50	60	7.5	3.9	1.1
12	45	Female	0.7	0.1	160	30	20	6.0	3.2	0.9
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	50	Male	4.2	2.3	450	69	50	7.0	3.0	0.7
2	42	Male	30.5	14.2	285	65	130	5.2	2.1	0.6
3	39	Male	6.6	3.0	215	190	950	4.0	1.7	0.7
4	19	Male	1.4	0.8	178	13	26	8.0	4.6	1.3
5	36	Male	1.7	0.5	205	36	34	7.1	3.9	1.2
6	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
7	29	Female	1.2	0.5	210	45	30	6.5	3.2	0.8
8	55	Male	3.4	1.5	400	78	90	7.0	3.8	1.1
9	34	Female	0.8	0.3	150	22	25	6.2	3.0	0.9
10	48	Male	2.5	1.0	320	50	60	6.8	3.5	1.0
11	41	Female	1.7	0.9	275	35	40	7.1	3.4	0.95
12	60	Male	0.9	0.2	180	20	28	5.9	2.8	0.7
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	28	Female	0.9	0.2	316	25	23	8.5	5.5	1.8
2	45	Male	3.2	1.4	512	50	58	6.0	2.7	0.8
3	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
4	30	Female	0.7	0.2	63	31	27	5.8	3.4	1.4
5	56	Female	0.7	0.1	145	26	23	7.0	4.0	1.3
6	36	Male	5.3	2.3	145	32	92	5.1	2.6	1.0
7	34	Female	1.2	0.4	210	30	28	6.5	3.1	1.05
8	29	Male	0.8	0.2	150	25	22	5.9	2.8	1.02
9	42	Female	2.5	1.0	300	45	55	7.2	3.5	0.85
10	50	Male	3.1	1.5	400	60	70	6.8	3.0	0.75
11	38	Female	1.0	0.3	180	35	40	6.0	2.9	0.95
12	45	Male	0.9	0.2	220	20	30	5.5	2.4	1.1
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	38	Male	1.7	0.7	859	89	48	6.0	3.0	1.0
2	55	Female	10.9	5.1	1350	48	57	6.4	2.3	0.5
3	60	Female	1.4	0.7	159	10	12	4.9	2.5	1.0
4	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4
5	40	Male	1.9	1.0	231	16	55	4.3	1.6	0.6
6	49	Female	0.8	0.2	158	19	15	6.6	3.6	1.2
7	34	Female	1.2	0.4	210	30	25	7.0	3.5	1.2
8	29	Male	0.9	0.3	180	22	19	6.5	3.0	1.1
9	45	Female	2.5	1.0	300	40	35	7.8	4.0	1.5
10	52	Male	3.1	1.5	400	50	45	8.2	3.8	1.0
11	60	Female	1.8	0.6	220	28	30	6.9	3.4	1.1
12	38	Male	2.0	0.8	250	35	40	7.5	3.7	1.3
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3